



NAN EI EI HTUN

Senior Software Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 9+ years of experience developing full-stack applications, REST APIs, and software solutions across healthcare, Biotechnology, banking, and automation. Strong expertise in C#/.NET, ASP.NET, SQL, PostgreSQL, React, TypeScript, Node.js, AWS, and Linux. In my current role at Vela Diagnostics, I develop and maintain laboratory software involving assay workflows, result interpretation, data visualization, reporting, data processing, instrument integration, and automation. I also have hands-on project experience with Python, machine learning, and AI-assisted development, with a strong focus on building reliable, maintainable, and user-focused software solutions.

TECHNICAL SKILLS

Programming & Frameworks	C#, .NET, .NET Core, Python, JavaScript, TypeScript, Node.js, React, Angular, Express.js, WPF, WCF, Autolt, LINQ, HTML5, CSS3, Bootstrap, jQuery, ASP.NET MVC
Databases	MSSQL, MySQL, PostgreSQL, Neo4j, Navicat
Architecture & Integration	Microservices, Domain-Driven Design (DDD), RabbitMQ, RESTful APIs, API Integration
DevOps & Tools	Git, Bitbucket, TFS, SVN, Jenkins, Octopus Deploy, TeamCity, Docker, PM2
Cloud & Infrastructure	AWS, Microsoft Azure, Linux (Ubuntu), Server Deployment and Management
Project Management Tools	Jira, Trello, Asana, Clockify
Reporting & Data	RDLC Reports, CSV/XML Data Processing
Testing & QA	Manual Testing, QA Collaboration, Bug Fixing, UAT support, Production support

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Vela Diagnostics PTE. LTD, Singapore

Aug 2023 – Present

- Developed and maintained C#, .NET/.NET Core, Web API, web and desktop applications supporting diagnostic laboratory workflows, including **SA201 Reporter**, **Vela Analytics**, **One QPCR**, **Sentosa Workflow Assistant**, **SX-Flex-Assistant**, **ID-List Generator** and **QuantStudio 12K Flex**.
- Developed backend business logic, REST APIs, data processing workflows, system integrations and application features for assay management and laboratory operations.
- SA201 Reporter**: Developed features for assay run management, **result interpretation**, **plate setup and data visualization**, **RDLC reporting**; implemented assay configuration logic including **CT value thresholds and positive/negative result determination**.
- Vela Analytics**: Deployed, configured and maintained the application on **Linux servers**, including services, dependencies, runtime environments and production support.
- One QPCR**: Enhanced diagnostic laboratory software with UI/UX improvements and automated data export functionality.
- Sentosa Workflow Assistant**: Implemented barcode validation to improve sample identification and workflow accuracy.
- ID-List Generator**: Developed plate setup layouts and XML generation functionality to support automated assay configuration and execution.

- ▶ **QuantStudio 12K Flex:** Developed **Autolt-based UI automation** to automate the end-to-end assay workflow from **SA201 Reporter to Thermo Fisher QuantStudio 12K Flex**, including launching the software, loading assay files and executing laboratory workflows.
- ▶ Developed CSV-to-XML conversion utilities and implemented cross-system enhancements including **Ant Design dark themes, PDF export and QR-code validation**.
- ▶ Performed software installation, configuration and troubleshooting, and production support for local laboratory deployments and internal testing environments.
- ▶ Contributed to a **qPCR machine-learning project using Python**, including data preparation and processing, model training/testing, evaluation, and working with a trained ML model.
- ▶ Created and maintained technical, functional and validation documentation, including **DCOs, CMRDs, FRDs, installation guides and test protocols**, supporting software traceability and quality processes.
- ▶ Supported annual audits by preparing and maintaining audit-ready documentation and validation records in accordance with established quality and compliance requirements.
- ▶ Collaborated with **PPS, QA and QC teams** to deliver features, support testing, troubleshoot software defects and ensure software quality and compliance.

Senior Software Developer

CB Bank, Myanmar

Apr 2022 – Jul 2023

- ▶ Collaborated on the development of the iBanking Platform for personal and corporate users.
- ▶ Integrated payroll system with scheduler and RabbitMQ for real-time message handling.
- ▶ Developed payment and transfer modules, customer service portals with account restriction and billing processes.
- ▶ Designed RDLC reports for dynamic financial reports including e-statements.
- ▶ Developed WCF web services, APIs and ensured cross-platform compatibility.
- ▶ Worked closely with QA teams during testing cycles to troubleshoot and resolve application defects.

Junior/Senior Software Developer

KoeKorTech IT Social Enterprise Co., Ltd, Myanmar

Apr 2016 – Mar 2022

- ▶ **MayMay / MayMay Plus:** Developed web app, Rest APIs, CMS and integrated Chatbot for Android app.
- ▶ **Central Statistical Organization (csostat.gov.mm):** Built website and CMS for national statistical system.
- ▶ **Christian Aid - Child Health Project:** Developed chatbot and CMS for vaccination and maternal health info.
- ▶ **KoeKoeTech Website:** Created frontend with reCAPTCHA integration and SMTP configuration.
- ▶ **Moe Thee Hospital Website:** Built frontend for a multidisciplinary hospital's official website.
- ▶ Collaborated with UI/UX designers, QA engineers and project stakeholders throughout the software development lifecycle

EDUCATION & CERTIFICATIONS

Bachelor's Degree in Computer Science

University of Computer Studies, Monywa, Myanmar

2011 – 2015

Diploma in Global English

University of Yangon, Myanmar

2018 – 2020

Project Management Certificate

Micro Max Education & Training Centre, Yangon, Myanmar

2021

PERSONAL STRENGTHS & INTERESTS

Strengths

- ▶ Excellent time management skills
- ▶ Strong team player and independent worker
- ▶ Hardworking, honest and detail-oriented
- ▶ Compliant with rules and procedures

Interests

- ▶ Technology and innovation
- ▶ Software architecture
- ▶ Continuous learning and self-improvement
- ▶ Project management and team leadership

